



UNIVERSITY OF SCIENCE - VNUHCM

FACULTY OF INFORMATION TECHNOLOGY

SOFTWARE TESTING

HW3 - SCENARIO TESTING

Software Testing Project Report

Authors:

Trần Thị Cát Tường
(22127444)

ttctuong22@clc.fitus.edu.vn

Supervisors:

Teacher Trần Duy Hoàng

Teacher Hồ Tuấn Thanh

Teacher Trương Phước Lộc

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1 Group Information

Group ID: 07

Member Name	Student ID	Assigned Features	Status
Cao Uyển Nhi	22127310	- SignUp	Done
		- Checkout	Done
Lưu Thanh Thuý	22127410	- SignIn	Done
		- User Management	Done
Nguyễn Phước Minh Trí	22127424	- Catalog	Done
		- Categories	Done
Võ Lê Việt Tú	22127435	- MyProfile	Done
		- Order Management	Done
Trần Thị Cát Tường	22127444	- Contact	Done
		- Category Management	Done

2 Scenario Definition

2.1 Feature 1: Contact (Guest User)

2.1.1 Flow

Basic Flow

1. The guest user accesses the Contact form page.
2. Enters First Name, Last Name, and Email.
3. Selects a Subject from the dropdown.
4. Writes a Message.
5. (Optional) Uploads an attachment file.
6. Clicks the **Submit** button.
7. The system displays a success message: “Message sent successfully.”

Alternate Flows

- **2.a.** First Name or Last Name or Email is empty → Inline error on blur, submission blocked.
- **2.b.** First Name or Last Name or Email exceeds 120 characters → Field rejected and error message displayed.
- **2.c.** Email is in invalid format (e.g., missing “@”, domain) → Error shown immediately.
- **3.a.** Subject is not changed from default value → Error on submit.
- **4.a.** Message is empty → Inline error shown when field is blurred.
- **4.b.** Message is shorter than 50 characters → Submit rejected with warning.
- **4.c.** Message is longer than 250 characters → Submit rejected with warning.
- **5.a.** File is uploaded with unsupported extension (e.g., .exe, .docx) → Upload blocked, error shown.
- **5.b.** File size exceeds 500KB → Upload blocked, error message shown.

2.1.2 Scenario Table

Scenario ID	Scenario Name	Starting Flow	Alternate Flow
SC-CON-01	Guest submits valid contact form	1 → 7	–
SC-CON-02	Unfulfilled prerequisites	Step 2	2.a
SC-CON-03	Email or Name exceeds 120 characters	Step 2	2.b
SC-CON-04	Email is in invalid format	Step 2	2.c
SC-CON-05	Subject is not selected from dropdown	Step 3	3.a
SC-CON-06	Message field is empty	Step 4	4.a
SC-CON-07	Message is shorter than 50 characters	Step 4	4.b
SC-CON-08	Message is longer than 250 characters	Step 4	4.c
SC-CON-09	File uploaded has unsupported extension	Step 5	5.a

Scenario ID	Scenario Name	Starting Flow	Alternate Flow
SC-CON-10	File uploaded exceeds 500KB	Step 5	5.b

2.2 Feature 2: Category Management (Admin)

2.2.1 Flow – Add Category

Basic Flow

1. In Category management Page, click button “Add Category”.
2. Selects **Parent ID** (optional).
3. Enters **Name** (no more than 120 characters, unique).
4. Enters **Slug** (lowercase, URL-safe, unique).
5. Clicks the **Save** button.
6. System shows success message: “Category created successfully.”

Alternate Flows

- **3.a.** Name field is left empty → Inline error on blur, submit blocked.
- **3.b.** Name exceeds 120 characters → Validation error shown.
- **3.c.** Name is not unique (already exists) → Error: “Name already exists.”
- **4.a.** Slug contains spaces, special characters → Error on submit.
- **4.b.** Slug field is empty → Submit blocked with error.
- **4.c.** Slug already exists in the database → Duplicate error shown.

2.2.2 Flow – Edit Category

Basic Flow

1. In Category Management page, click “Edit” on a category row.
2. Select Parent ID (optional).
3. Edit Name (required, no more than 120 chars, unique).
4. Edit Slug (required, lowercase, hyphenated, unique).
5. Click Save.
6. System displays success message: “Category updated successfully.”

Alternate Flows

- **3.a.** Name field is empty → validation error.
- **3.b.** Name exceeds 120 characters → error shown.
- **3.c.** Name duplicates another category’s name → error shown.
- **4.a.** Slug is empty → validation error.
- **4.b.** Slug contains spaces, special characters, or uppercase → format error.
- **4.c.** Slug already exists (conflicts with another category) → duplicate error.

2.2.3 Flow – Search Category

Basic Flow

1. In Category Management page, enter a keyword in the search bar.
2. System filters category list by name (partial match, case-insensitive).

3. Matching results are displayed.

Alternate Flows

- **1.a.** Keyword matches no categories → system shows empty list.
- **1.b.** Keyword field is left blank → system shows full category list.

2.2.4 Flow – Delete Category

Basic Flow

1. In Category Management page, click “Delete” on a category.
2. System deletes category and shows message: “Category deleted successfully.”

Alternate Flows

- **1.a.** Category is parent of another category → error: “This category is used elsewhere.”
- **1.b.** Category contains products → error: “This category contains products and cannot be deleted.”

2.2.5 Scenario Table

Add category

Scenario ID	Scenario Name	Starting Flow	Alternate Flow
SC-CAT-01	Successfully add a new category	1 → 6	–
SC-CAT-02	Name field is left empty	Step 3	3.a
SC-CAT-03	Name exceeds 120 characters	Step 3	3.b
SC-CAT-04	Name is not unique	Step 3	3.c
SC-CAT-05	Slug contains invalid characters	Step 4	4.a
SC-CAT-06	Slug field is empty	Step 4	4.b
SC-CAT-07	Slug already exists in the database	Step 4	4.c

Edit category

Scenario ID	Scenario Name	Starting Flow	Alternate Flow
SC-CAT-EDIT-01	Edit category successfully	1 → 6	–
SC-CAT-EDIT-02	Parent ID is same as category’s own ID	Step 2	2.a
SC-CAT-EDIT-03	Name is empty	Step 3	3.a
SC-CAT-EDIT-04	Name exceeds max length	Step 3	3.b
SC-CAT-EDIT-05	Name already exists	Step 3	3.c

Scenario ID	Scenario Name	Starting Flow	Alternate Flow
SC-CAT-EDIT-06	Slug is empty	Step 4	4.a
SC-CAT-EDIT-07	Slug is not URL-safe (e.g., spaces/uppercase)	Step 4	4.b
SC-CAT-EDIT-08	Slug already exists	Step 4	4.c

Search category

Scenario ID	Scenario Name	Starting Flow	Alternate Flow
SC-CAT-SEARCH-01	Search with partial name match	1 → 3	–
SC-CAT-SEARCH-02	Search with no matching result	Step 1	1.a
SC-CAT-SEARCH-03	Search with empty keyword	Step 1	1.b

Delete Category

Scenario ID	Scenario Name	Starting Flow	Alternate Flow
SC-CAT-DEL-01	Successfully delete a category	1 → 3	–
SC-CAT-DEL-02	Category is being used as Parent	Step 1	1.a
SC-CAT-DEL-03	Category contains products	Step 1	1.b

3 Use of AI Tool

3.1 Tool name

ChatGPT provided by OpenAI.

3.2 Prompt used:

You are a professional software tester. Based on the feature description and the scenario table below, generate ****detailed manual test cases**** in QA-standard format and ****output them in a format suitable for Excel****.

Feature: [Feature name]

Basic Flow:
[Basic Flow]

Alternate Flows:
[Alternate Flows]

Scenario Table:
[Scenario Table]

For each test case, include the following columns:

- * Test Case ID (e.g., TC01)
- * Title
- * Preconditions (if any)
- * Inputs (Test Data)
- * Test Steps
- * Expected Result
- * Actual Result (leave blank)
- * Result (Pass/Fail - leave blank)

3.3 AI-generated vs Manually Refined Test Cases

- **AI-generated test cases:** The AI tool generated **one basic test case per scenario**, focusing on the general validation logic. For example:
 - A single test case was created to cover all cases where First Name, Last Name, or Email were missing — grouped as one.
 - The AI also used generic descriptions such as “invalid slug” or “invalid file” without specifying the exact nature of the invalid input (e.g., containing special characters, uppercase letters, or empty value).
 - Each test case only addressed **a high-level violation** of the expected constraints and did not explore detailed boundary or alternate conditions.
- **Manually created & refined test cases:** I reviewed the AI-generated test cases and made the following manual improvements:

- **Split generalized validations into separate, specific test cases.** For example:
 - * I created distinct test cases for missing First Name, missing Last Name, and missing Email — instead of grouping them.
 - * For the “invalid slug” case, I created different test cases for each invalid condition: containing spaces, uppercase letters, or being empty.
- **Added test cases that were missing** in the AI results:
 - * The valid/happy-path scenario where the form is correctly submitted.
 - * Edge cases involving multiple simultaneous validation failures (e.g., invalid email + short message + no subject).
 - * Backend/server-related failures and message feedback upon submission.

These manually created cases ensured broader test coverage, more accurate mapping to real-world behavior, and better alignment with scenario testing goals.

4 Self-Evaluation (Scenario Testing)

Criteria	Self-Evaluation	Notes
At least 2 Scenario Selection	1.0 / 1.0	Selected two relevant and realistic user scenarios from The Toolshop system.
Scenario Testing	2.0 / 2.0	Clearly described scenarios and applied scenario-based thinking.
Use of AI Tools	1.0 / 1.0	Stated tool, prompts used, validation process, and usage scope.
Test Execution	0.5 / 0.5	Executed all designed test cases locally and documented results.
Bug Reporting	0.5 / 0.5	Reported bugs with full reproduction steps and severity analysis.
Scenario Testing Report	1.0 / 1.0	Report is clear and traceable